Jonathan Tan

Data Science & Analytics Lead

+61450787453
jtsw1990@gmail.com
jtsw1990
https://thesourceco.de
Sydney, NSW

Skills & Stack

Data Strategy, System Design, Data Architecture, Statistical modelling, Pricing, Financial modelling, Data visualisation, Data analytics, Machine learning, MLOps, CICD, Reporting, A/B Testing, Web scraping, AI Engineering

Python, R, Excel, Vba, SQL, Snowflake, Databricks, PySpark, Buildkite, AWS, Azure Cloud, Git, Javascript, NodeJS, Postman, React, NGINX, MongoDB, JIRA, Confluence, PowerBI, Tableau

Education

2015-2017 M. Actuarial Science
Australian National University

2012-2014 B. Accountancy
Royal Melbourne Institute of Technology

Awards & Certifications

2025 SCOR GemStar award winner
2023 SCOR Iron Viz winner
2022 nib FACS recognition winner
2018 Part 1 & 2 Actuarial papers
2014 Awarded CPA scholarship

Publications

2023 SCOR Iron Viz Tableau Submission – Small Changes, Big Impact

2021 Actuaries digital – Explainable ML: A peek into the black box through SHAP

2019 Actuaries digital – Version control, because the Recycle Bin doesn't count

2018 Actuaries digital – Natural Language Processing, text classification

Experienced data leader and actuary with a track record of building and executing data strategy, leading cross-functional analytics teams, and delivering end-to-end data products. Equally comfortable shaping data strategy at the executive level and rolling up my sleeves to build solutions, bringing both strategic direction and hands-on technical depth.

Experience

Data Science & Analytics Lead @ SCOR SE
Jun 2023 – Present

- Represented ANZ in the global Tableau Viz Games, winning first place on the sustainability challenge, and winning Viz of the day on Tableau
- Designed and developed the ANZ bordereaux processing engine, reducing manual effort by 70%. Implemented data schema and model, creating over 500+ validations and data quality tracking
- Designed and co-contributed to local data governance framework including development best practices, policies, risk matrices, RACI and stewardship model, aligned with general CPS framework
- Steered company data strategy alongside executive team:
  - Built and design repository templating system for data projects on git, working with I.T. Includes dependency management, linting, CICD, test suites and structured documentation & cloud deployments (MS Azure stack)
  - Designed fit-for-purpose data layers and E(t)LT pipelines for bordereaux data for downstream financial & regulatory reporting
  - Lead project for onboarding local data systems and processes onto global platforms such as Databricks & Palantir, and advised on budget reduction with smart use of licenses (saving > $250k AUD annually)
  - Provided direction and commentary on board communications regarding local data strategy and progress
- Led ANZ data standardisation plan for global project across more than 50 tables
- Collaborated with the Valuation team to designed and built data pipelines & processes for APRA reporting on IDII and decline rates
- Worked with claims & I.T teams (internal & external) to understand client (PAS, CMS, finance) systems such as Fineos, IMars, Solvexia
- Designed and developed A.I assisted process (including validation framework) to map free text claim causes to ICD10 codes across whole portfolio, reducing manual work by 90%
- Led and developed a team of 3 analysts, including resourcing across different functions (EA, data, tenders) while providing coaching to actuarial teams on data science methods, tooling and AI usage. Also providing thought leadership articles to clients & industry on data visualisation and other techniques/tools.
- Led research and implementation of GI statistical modelling methods for group life pricing, spanning CL triangles, GLMs. Tender work with client dashboards using geospatial risk heatmaps (postcodes & SA4 where available)

Manager – Data & Insights @ nib Health Funds
Oct 2022 – Jun 2023

- Led a cross-functional data taskforce that produced analytics and reporting enabling resolution of 5,000+ GICOP compliance breaches within 3 months
- Steered decision for global modelling system, working with procurement, I.T and inhouse stakeholders to design requirements and compare costings and managing existing vendor & consultant relationships
- Developed company-wide data science competency through "Analytics & Insights" Workplace page and "Kernel Academy" – a machine learning knowledge sharing group
- Actuarial & analytics SME for onboarding current data process and systems onto client systems like Amadeus into local data model
- Led and developed systems and data integration from M&A with QBE's travel portfolio and PAS (Pegasus) into local policy system and data model
- Built and design dbt pipeline on Snowflake, working with I.T to manage team repositories, permissions and version control processes
- Managed and mentored a team of 4 junior analysts to deliver data products across bordereaux reporting, customer and claims analytics, operational analytics

Data Science Lead @ thecanary.xyz
May 2022 – Oct 2022

- Established data team structure of 5 and development processes for an MVP build in a fast-moving cryptocurrency start-up environment
- Led decisions on selection and purchase for systems spanning: VCS, source APIs, cloud services (storage, compute & hosting, AWS stack)

- Designed and built backend ETL architecture and analytics API across multiple source systems using FastAPI, EC2, Postgres, and Python

Senior Data Scientist @ nib Health Funds

May 2021 – Oct 2022

- Developed and implemented COVID pricing strategy through Delta and Omicron waves resulting in a 40% improvement to portfolio performance
- Lead development in end-to-end inhouse technical travel insurance pricing models
- Delivered commercial models for multiple RFPs totalling an estimated $80m GWP
- Developed data science strategy and model architecture for actuarial team
- Designed and developed bordereaux automation process, worked with I.T to integrate local scripts to SAP (BO) through Raylight (SDK)

Pricing Engine Lead @ IAG Ltd

May 2020 – May 2021

- Lead development and launch of Caravan, Trailer, Strata portfolios worth $50m GWP
- Sprint planning and resource allocation for a team of 4 under AGILE framework, triaging and planning work using JIRA
- Built pipeline to automate rate setting process by 60% using Python/R scripting
- Collaborated with business API team to develop ingestion schema for conversion between flat and JSON data
- Built and design automated testing framework for local pricing team using Splunk logs and APIs into central scripts
- Lead deployment workflow and integration testing for Home/Motor pricing algorithms using Postman and R

Actuarial Analyst @ nib Health Funds

July 2018 – May 2020

- Developed earnings model with MLE and goodness of fit tests using Scipy/Statsmodels
- Built and deployed travel alerts and embargo decision chatbot using Slack RTM API, connecting multiple systems into a simple program (Slack RTM, Snowflake, local OLAP cubes)
- Developed and maintained recurring ETL processes in Snowflake/ Python for reporting
- Modelled and launched new rating factor for direct brand worth $120m GWP annually and improved portfolio performance by 10%

Other interests/ projects

2025 Started Code&Grow for kids and volunteering at BPS
2024 Co-founder & CTO of data consultancy Bastion Labs Pty Ltd
2023 Developed AI content pipeline “Glimpse” on Instagram
2022 Developed an NFT project with an open-source team
2022 Full stack insurance pricing engine built from scratch in Python/Flask/Nodejs/React